

Andrew Balbuena

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EDUCATION

University of California, San Diego

B.S. Mechanical Engineering

La Jolla, CA

Expected Graduation: June 2028

EXPERIENCE

UCSD TEMP Laboratory (PI: Renkun Chen)

Engineering Research Undergraduate

La Jolla, CA

01 October 2025 - Present

- Supported **design and testing** of **fiber-membrane cooling** concepts for **high-power electronics**; contributed to **thin-film evaporation** experiments including **prototype fabrication**, **thermal data collection**, and **performance characterization**
- Designed and manufactured **test fixtures** and **custom experimental hardware** using **Autodesk Fusion (CAD/CAM)**, **CNC milling**, **laser cutting**, and **3D printing**; built enabling components (e.g., **PEEK clamp**, **custom gaskets**, **machined heating block**) to improve sealing, alignment, and controlled thermal input for repeatable runs
- Analyzed and visualized experimental datasets in **MATLAB** and **Excel** (cleaning, trend extraction, and comparison across trials); iterated fixtures to improve **measurement consistency** and stabilize test conditions for more reliable thermal/force-related results

City of Delano Recreation

Youth Program Leader

Delano, CA

02 December 2022 – July 2025

- Led and supervised daily programming for **~15 youth (ages 7–17)**, coordinating activities and maintaining a **safe, structured environment**; strengthened **leadership**, **communication**, and **team coordination** through real-time decision-making and conflict resolution
- Planned and executed** schedules, set-up/tear-down, and resource organization for recurring events; improved **operational efficiency** by standardizing routines and **adapting** plans to staffing, space, and equipment constraints
- Supported facility and equipment upkeep through hands-on **maintenance**, troubleshooting, and basic repairs; applied a **process-improvement mindset** to reduce recurring issues and keep programs running reliably

PROJECTS

Four-Wheel Drive Robot (Robot Competition)

Fall 2025

- Researched, designed, and manufactured a four-wheel drive competition robot to reliably score soccer balls in **60-second head-to-head matches** while meeting strict **size, motor, and material constraints**
- Used **AutoCAD (2D)** and **Inventor (3D)** for CAD and assembly; applied **laser cutting** and shop processes (**drilling, tapping, reaming, press fits**) to fabricate and iterate the drivetrain and scoring mechanism; validated motion using CAD simulation
- Achieved **8 goals in <30 seconds** (no opponent) and demonstrated repeatability across **4/4 test runs**; in competition, played **5 matches** and consistently scored **~5 goals per match** against opponents

PEEK Clamp (Precision Machining / Fixturing)

Fall 2025

- Designed and manufactured a **PEEK clamp** to securely retain experimental hardware while maintaining alignment and preventing damage to sensitive components during lab testing
- Modeled in **Autodesk Fusion**, then machined on a **manual mill** using **edge finding** to establish a datum and the mill's **DRO/coordinate monitor (X/Y/Z)** to accurately locate features; held ± 0.05 mm hole-to-spacing accuracy so **all hole patterns aligned** during assembly
- Produced a repeatable, tight-fit clamp that improved **setup consistency** and reduced misalignment issues during experimental trials by ensuring reliable part location and clamping

Custom Gaskets (Process Improvement / High-Flow Water Sealing)

Fall 2025

- Developed a repeatable process to fabricate **custom gaskets** for experimental assemblies requiring **leak-free sealing** during **high-flow water testing** and frequent rebuilds
- Designed gasket geometries/templates in **Fusion/AutoCAD** and standardized cutting/assembly steps to improve repeatability, reduce rework, and speed turnaround between test iterations
- Reduced gasket manufacturing time from **~1.5 hours to ~20 minutes per gasket** (excluding curing) and achieved reliable **leak-free sealing** during water tests up to 2 m/s across months of iterative experimentations.

SKILLS

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|--------------|----------|----------------------------|-----------------|
| • SolidWorks | • Excel | • Design for Manufacturing | • AutoCAD |
| • Fusion 360 | • MATLAB | • Shop Processes | • CNC Machining |